aiPHeed: food-insecurity risk indicators for CALABARZON (DA Region IV-A)

Information cutoff 2026-10-14T23:59:59+08:00. Record SHA-256 3ba6c184a0ba024ebbfc678add162fcff609ea5462fe6952c76d033b6e863af5.
aiPHeed produces potential food-insecurity risk indicators for review. They are not official food-insecurity classifications and do not replace assessments by authorized government agencies.

What is estimated

T1: S1 = H + 2D > tau, HIGH or LOW (the approved Chapter I composite food-stress label)
H: SWS Balance Luzon self-rated hunger (%), one regional value for all five provinces (measured)
D: province food inflation minus the five-province mean (pp), PSA provincial CPI (measured)
Not: an official or measured food-insecurity prevalence; provincial differences come only from prices

CALABARZON: official regional hunger (measured)

Quarter	Round	Hunger %	Posted
2024-Q2	Second Quarter 2024 SWS Survey, June 23-July	19.6	2024-07-23
2024-Q3	September 2024 SWS Survey	18.1	2024-10-17
2024-Q4	December 2024 SWS Survey	25.3	2025-01-14
2025-Q1	Stratbase-SWS March 15-20, 2025 National Surv	24.0	2025-03-29
2025-Q4	November 2025 SWS Survey	16.7	2026-01-14
2026-Q1	March 2026 SWS Survey	22.4	2026-05-12
2026-Q2	June 2026 SWS Survey	17.3	2026-08-24
2026-Q3	September 2026 SWS Survey	23.6	2026-10-14

regional: ONE value shared by all five provinces; not a provincial or municipal measurement

2026-Q3: first unseen quarter (frozen forecast vs released outcome)

Province	P(HIGH) frozen 1 Jul	Observed S1	Observed label
Cavite	73%	24.5467	HIGH

Laguna	74%	24.1467	HIGH
Batangas	63%	18.3467	HIGH
Rizal	72%	25.0133	HIGH
Quezon	63%	25.9467	HIGH

Source	Arm	Lead	Accuracy	Brier
E24v2 SEAL_20261013T164218Z (g	B1_naive_persistence	1	0.8	0.2
E24v2 SEAL_20261013T164218Z (g	B1p_calibrated_persistence	1	0.8	0.1389
E24v2 SEAL_20261013T164218Z (g	FINAL_V2_PCF	1	1.0	0.0963
E24v2 SEAL_20261013T164218Z (g	M2_lgbm_P_E24frozen	1	1.0	0.0375
E24v2 SEAL_20261013T164218Z (g	PCF_S_supplementary	1	1.0	0.1028
E31_FROZEN_Q3 (git 70d8770)	PCF21	1	1.0	0.0974
E31_FROZEN_Q3 (git 70d8770)	B1_persistence	1	1.0	0.0
E31_FROZEN_Q3 (git 70d8770)	B1p_calibrated_persistence	1	1.0	0.0467
E31_FROZEN_Q3 (git 70d8770)	CLIM_climatology	1	1.0	0.2221
E31_FROZEN_Q3 (git 70d8770)	ALWAYS_HIGH	1	1.0	0.0
E31_FROZEN_Q3 (git 70d8770)	PCF21	2	1.0	0.1002
E31_FROZEN_Q3 (git 70d8770)	B1_persistence	2	0.8	0.2
E31_FROZEN_Q3 (git 70d8770)	B1p_calibrated_persistence	2	0.8	0.1379
E31_FROZEN_Q3 (git 70d8770)	CLIM_climatology	2	1.0	0.25
E31_FROZEN_Q3 (git 70d8770)	ALWAYS_HIGH	2	1.0	0.0
E31_FROZEN_Q3_P45_LEAD2 (git b	P45	2	0.6	0.1942
E31_FROZEN_Q3_P45_LEAD2 (git b	F12	2	1.0	0.0934
E31_FROZEN_Q3_P45_LEAD2 (git b	F12_SWS	2	1.0	0.1052
E31_FROZEN_Q3_P45_LEAD2 (git b	B2_lr_combined	2	1.0	0.0007

E31_FROZEN_Q3_P45_LEAD2 (git b	B3_lr_gov_only	2	1.0	0.0003
E31_FROZEN_Q3_P45_LEAD2 (git b	B4_lr_nlp_only	2	0.8	0.1885
E31_FROZEN_Q3_P45_LEAD2 (git b	B5_lgbm_gov_only	2	1.0	0.0398
E31_FROZEN_Q3_P45_LEAD2 (git b	B6_lgbm_nlp_only	2	0.6	0.1888

ONE quarter, five rows that share one regional H: a single observation of regional hunger. No statistical claim of skill can rest on it; it is the first unseen-quarter data point.

Cavite

	2026-Q4 NOWCAST (lead 0)	2027-Q1 FORECAST (lead 1)
P(S1 above threshold)	96%	91%
Signal	STRONG	STRONG
S1 80% interval	[17.5457, 28.4398]	[15.9363, 29.6972]
Regional hunger (SWS Balance Luzon) relative to the threshold	+6.82	+6.82
Provincial relative food-price pressure (2 x D)	+0.70	+0.52

exact additive decomposition of the estimate's margin over the threshold, S1 - tau = (H - tau) + 2D; the model's inputs, not causes
Food inflation in Cavite was 0.47 pp above the five-province average in 2026-Q3 (PSA, measured).
News 2026-Q4 (1-14 Oct): 0 validated article(s) of 13 province-attributed; status NO_USABLE_EVIDENCE.
Review: the estimate is high mainly because regional hunger is high; check provincial market prices and field reports. Prioritise for review by DA Region IV-A; do not allocate resources on this estimate alone.

Top LGUs (indicative)	Risk index	Poverty value kind
Gen. Mariano Alvarez	0.6937	province-mean substitute (no municipal e
Rosario	0.6133	province-mean substitute (no municipal e
Tagaytay	0.5996	PSA municipal SAE estimate
Cavite City	0.5003	province-mean substitute (no municipal e
Trece Martires	0.4183	province-mean substitute (no municipal e

Laguna

	2026-Q4 NOWCAST (lead 0)	2027-Q1 FORECAST (lead 1)
P(S1 above threshold)	96%	91%
Signal	STRONG	STRONG
S1 80% interval	[17.2511, 28.1452]	[15.7161, 29.477]
Regional hunger (SWS Balance Luzon) relative to the threshold	+6.82	+6.82

Provincial relative food-price pressure (2 x D)	+0.40	+0.30

exact additive decomposition of the estimate's margin over the threshold, S1 - tau = (H - tau) + 2D; the model's inputs, not causes
Food inflation in Laguna was 0.27 pp above the five-province average in 2026-Q3 (PSA, measured).
News 2026-Q4 (1-14 Oct): 0 validated article(s) of 8 province-attributed; status NO_USABLE_EVIDENCE.
Review: the estimate is high mainly because regional hunger is high; check provincial market prices and field reports. Prioritise for review by DA Region IV-A; do not allocate resources on this estimate alone.

Top LGUs (indicative)	Risk index	Poverty value kind
San Pablo	0.611	PSA municipal SAE estimate
San Pedro	0.5951	province-mean substitute (no municipal e
Biñan	0.4973	province-mean substitute (no municipal e
Cabuyao	0.443	province-mean substitute (no municipal e
Santa Cruz	0.3006	province-mean substitute (no municipal e

Batangas

	2026-Q4 NOWCAST (lead 0)	2027-Q1 FORECAST (lead 1)
P(S1 above threshold)	76%	77%
Signal	MODERATE	MODERATE
S1 80% interval	[12.9795, 23.8736]	[12.5228, 26.2838]
Regional hunger (SWS Balance Luzon) relative to the threshold	+6.82	+6.82
Provincial relative food-price pressure (2 x D)	-3.87	-2.89

exact additive decomposition of the estimate's margin over the threshold, S1 - tau = (H - tau) + 2D; the model's inputs, not causes
Food inflation in Batangas was 2.63 pp below the five-province average in 2026-Q3 (PSA, measured).
News 2026-Q4 (1-14 Oct): 2 validated article(s) of 17 province-attributed; status LIMITED_SIGNAL.
Monitor: the estimate is above the threshold with less certainty. Prioritise for review by DA Region IV-A; do not allocate resources on this estimate alone.

Top LGUs (indicative)	Risk index	Poverty value kind
Sto. Tomas	0.5136	province-mean substitute (no municipal e

Nasugbu	0.4917	PSA municipal SAE estimate
Taal	0.4806	province-mean substitute (no municipal e
Malvar	0.4638	province-mean substitute (no municipal e
Tanauan	0.443	province-mean substitute (no municipal e

Rizal

	2026-Q4 NOWCAST (lead 0)	2027-Q1 FORECAST (lead 1)
P(S1 above threshold)	97%	92%
Signal	STRONG	STRONG
S1 80% interval	[17.8894, 28.7835]	[16.1932, 29.9542]
Regional hunger (SWS Balance Luzon) relative to the threshold	+6.82	+6.82
Provincial relative food-price pressure (2 x D)	+1.04	+0.78

exact additive decomposition of the estimate's margin over the threshold, S1 - tau = (H - tau) + 2D; the model's inputs, not causes
Food inflation in Rizal was 0.71 pp above the five-province average in 2026-Q3 (PSA, measured).
News 2026-Q4 (1-14 Oct): 0 validated article(s) of 7 province-attributed; status NO_USABLE_EVIDENCE.
Review: the estimate is high mainly because regional hunger is high; check provincial market prices and field reports. Prioritise for review by DA Region IV-A; do not allocate resources on this estimate alone.

Top LGUs (indicative)	Risk index	Poverty value kind
Rodriguez	0.659	PSA municipal SAE estimate
Taytay	0.4399	province-mean substitute (no municipal e
Cainta	0.3901	province-mean substitute (no municipal e
Angono	0.2317	province-mean substitute (no municipal e
San Mateo	0.231	province-mean substitute (no municipal e

Quezon

	2026-Q4 NOWCAST (lead 0)	2027-Q1 FORECAST (lead 1)
P(S1 above threshold)	98%	93%
Signal	STRONG	STRONG
S1 80% interval	[18.5768, 29.4709]	[16.7071, 30.468]
Regional hunger (SWS Balance Luzon) relative to the threshold	+6.82	+6.82
Provincial relative food-price pressure (2 x D)	+1.73	+1.29

exact additive decomposition of the estimate's margin over the threshold, S1 - tau = (H - tau) + 2D; the model's inputs, not causes
Food inflation in Quezon was 1.17 pp above the five-province average in 2026-Q3 (PSA, measured).
News 2026-Q4 (1-14 Oct): 1 validated article(s) of 10 province-attributed; status LIMITED_SIGNAL.
Review: the estimate is high mainly because regional hunger is high; check provincial market prices and field reports. Prioritise for review by DA Region IV-A; do not allocate resources on this estimate alone.

Top LGUs (indicative)	Risk index	Poverty value kind
Sariaya	0.6612	PSA municipal SAE estimate
Tiaong	0.5306	province-mean substitute (no municipal e
San Antonio	0.5263	province-mean substitute (no municipal e
Plaridel	0.5231	province-mean substitute (no municipal e
Dolores	0.5193	province-mean substitute (no municipal e

Performance metrics (development quarters, lead 1)

Arm	n	HIGH/LOW	Acc	Bal.acc	Prec	Recall	F1	ROC-AUC	Brier
Naive persistence (Objective 4-I)	45	41/4	0.889	0.826	0.974	0.902	0.937	0.826	0.111
Always HIGH (majority class)	45	41/4	0.911	0.500	0.911	1.000	0.953	0.500	0.089
PSA-only indicator baseline (Objective 4-II)	45	41/4	0.600	0.668	0.960	0.585	0.727	0.756	0.280
NLP-only sentiment baseline (4-III)	45	41/4	0.267	0.146	0.750	0.293	0.421	0.122	0.309
Logistic regression baseline (4-IV)	45	41/4	0.511	0.619	0.952	0.488	0.645	0.652	0.308
Proposed LightGBM, E28 inputs	45	41/4	0.533	0.631	0.955	0.512	0.667	0.720	0.312
P45: proposed LightGBM, 45 candidates (resea	40	37/3	0.825	0.752	0.969	0.838	0.899	0.658	0.133
PCF21 (served engine)	45	41/4	0.867	0.701	0.949	0.902	0.925	0.762	0.103

Several arms exceed 75% accuracy at lead 1, but so does always-HIGH (0.911) because 91% of test labels are HIGH. No arm beats persistence AND always-HIGH on accuracy or Brier, so accuracy above 75% does not show forecasting skill here; the requirement is met only in a form that a constant forecast also meets.

News relevance validation (blind holdout)

Rule R1: precision 0.852 (95% CI [0.773, 0.92]), recall 0.395 (95% CI [0.263, 0.693]), F1 0.54.
Data sufficiency (news, FSSI-v2): {"collected": 8320, "province_attributed": 2820, "relevant_R1": 140, "syndicated_copies_removed": 0, "admitted_to_FSSI_v2": 140, "province_quarters_sufficient_v2": 7, "province_quarters_limited_v2": 46, "province_quarters_none_v2": 67, "in_FSSI_v1": 55, "province_quarters_sufficient_v1": 0}

Limitations

- The label is 68% regional SWS hunger: one survey value drives all five provinces, so most quarters (and 2026-Q3) are HIGH everywhere. Provincial differences come only from relative food prices.
- No lead time is validated: on development quarters no model beats persistence and always-HIGH; the unseen 2026-Q3 is one regional observation (five rows that share one H).
- News evidence is title-only. The relevance rule R1 has precision 0.85 and estimated recall 0.40 on a blind holdout (deployed R1.1 fixes its false-positive patterns; not yet re-validated on fresh data), so most province-quarters still have limited or no news evidence.
- 126 of 142 LGUs use a province-mean poverty substitute (2021 municipal SAE extract not available without loading pre-2021 values); their order inside a province comes from population density.

- Annotation of the relevance samples was done by the AI assistant, not by the researchers; a researcher re-review of validation/*/annotation_sheet_blind.json is recommended before thesis use.
- Google News robots.txt disallows automated access to /rss for generic crawlers; continued use of this source needs the researcher's decision (see E31 FINAL_REPORT).

aiPHeed produces potential food-insecurity risk indicators for review. They are not official food-insecurity classifications and do not replace assessments by authorized government agencies.